

To: Sam
From: Abubaker Ahmadi
Date: August 14, 2026
Subject: Book Scraper as an ETL Pipeline

Hi Sam,

Here is how the book scraper works as an ETL pipeline. The three stages are split across scraper.py and main.py.

Extract

scraper.py fetches the site. scrape_all_books() walks every category listed in the nav, scrape_category() follows pagination within a category, and scrape_book() fetches each individual product page.

Transform

Also done in scraper.py: each call to scrape_book() parses the fetched HTML into one flat dictionary per book, with keys for product_page_url, universal_product_code, book_title, price_including_tax, price_excluding_tax, quantity_available, product_description, category, review_rating, and image_url.

Load

main.py runs the Load stage. It takes the list of book dictionaries returned by scrape_all_books() and writes them straight to data/book_data.csv using csv.DictWriter, with one row per book and a fixed column order matching the dictionary keys. Separately, it downloads each book's cover image and saves it under images/<category>/, using the site's own filename.

Summary

scraper.py handles Extract and Transform, and main.py handles Load: one CSV file for all the book data, plus a folder of downloaded cover images.

Best,
Abubaker Ahmadi